Chess v1 — Cut Plan

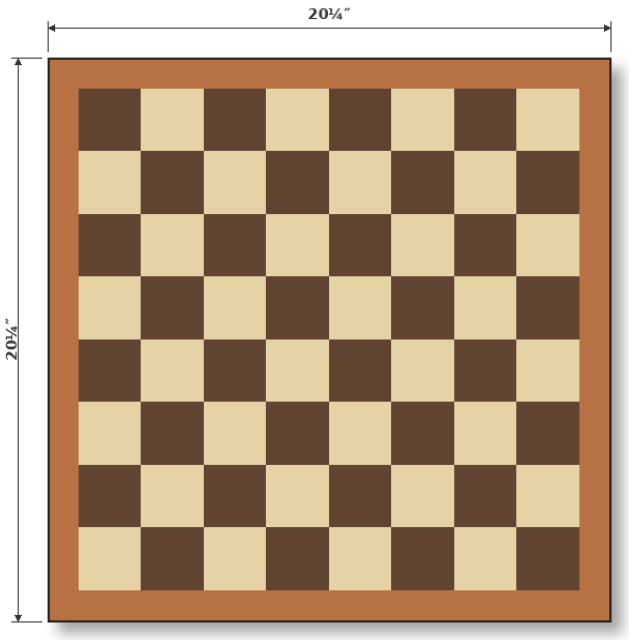
One end-grain chess-pattern board, walnut and brown maple 2.25 in cells, cherry border 1.125 in all sides, finished 20.25 in square, 0.5 in thick. Each row is built as two half-width panels split at the centre line (left half mirrors right half), so no panel is wider than 10.125 in and every panel fits the 12.5 in planer. Only three panel patterns are built, A, B and C, and both A and B start with a cherry component. Their mirrors are the same slices placed flipped (flip H), A flipped covers the old BR half, B flipped covers the old AR half, C flipped covers the old CR half.

This is a generated plan: the finished design and the cut list that realizes it on the boards in hand. Where the cut list cannot be met from inventory it calls for **bought boards** — buy those and re-run before cutting. The sections follow the build order, from the first crosscut to the final slice-grid assembly.

1. Warnings

#	Lvl	Type	Message
1	WARNING	Small Margin	Board BW1A leaves only ⅛″ margin (< ¼″) — too tight; prone to under-supplying strips if the board runs narrow.
2	WARNING	Zero Offcut	6 strip(s) crosscut to an exact fit with 0″ offcut — no margin for cutting error in the crosscut phase: BM1A ×2, BC1A ×1, BC2A ×1, BW1A ×2. A cut even slightly long, or kerf drift, will make the last piece short. Consider longer blanks or fewer pieces per strip.

2. Finished board pattern



3. Volume Sanity Check

A per-species feasibility check in board feet (1 BF = 144 cubic inches). **Required** is the surfaced volume of every installed slice across all 1 board(s) plus the volume that must be removed to produce them — flattening (sliceDepth − finishedThickness), planing all four edges (minPlaneLoss), and one miter kerf per slice. **Available** is the inventory stock volume. **Volume Offset** = available − required, with available as a percentage of required in parentheses (100% means they are equal); a negative value

means the project cannot be built from stock in hand.

Species	Available	Required	Volume Offset
maple	4.7 BF	0.6 BF + 1.5 BF loss = 2.0 BF	+2.7 BF (235%)
cherry	2.6 BF	0.3 BF + 0.8 BF loss = 1.1 BF	+1.5 BF (231%)
walnut	2.7 BF	0.6 BF + 1.5 BF loss = 2.0 BF	+0.7 BF (134%)

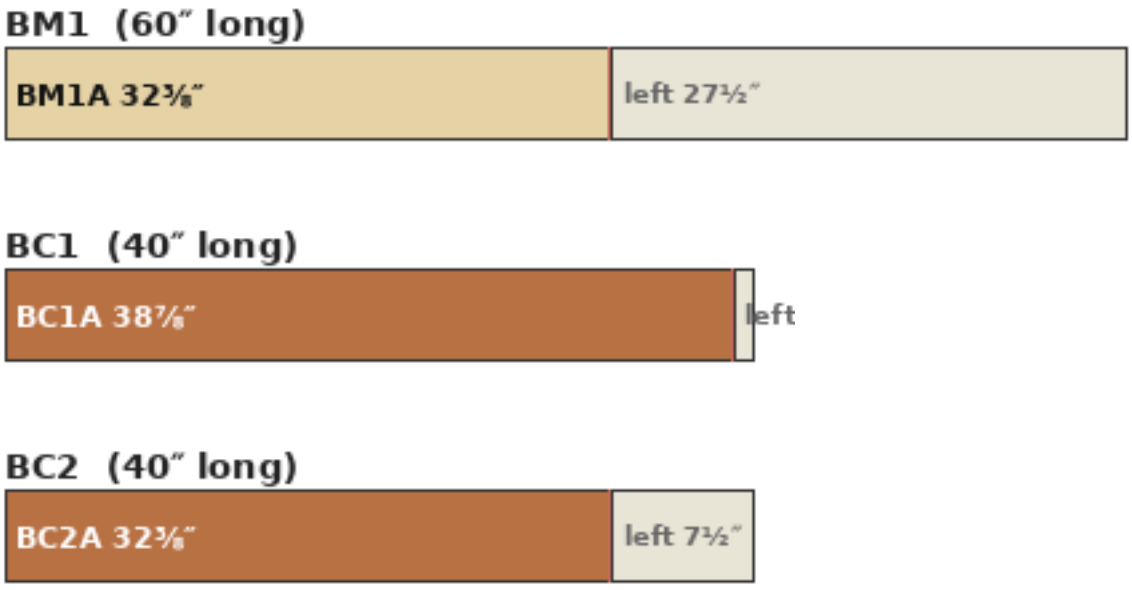
4. Crosscut Blanks

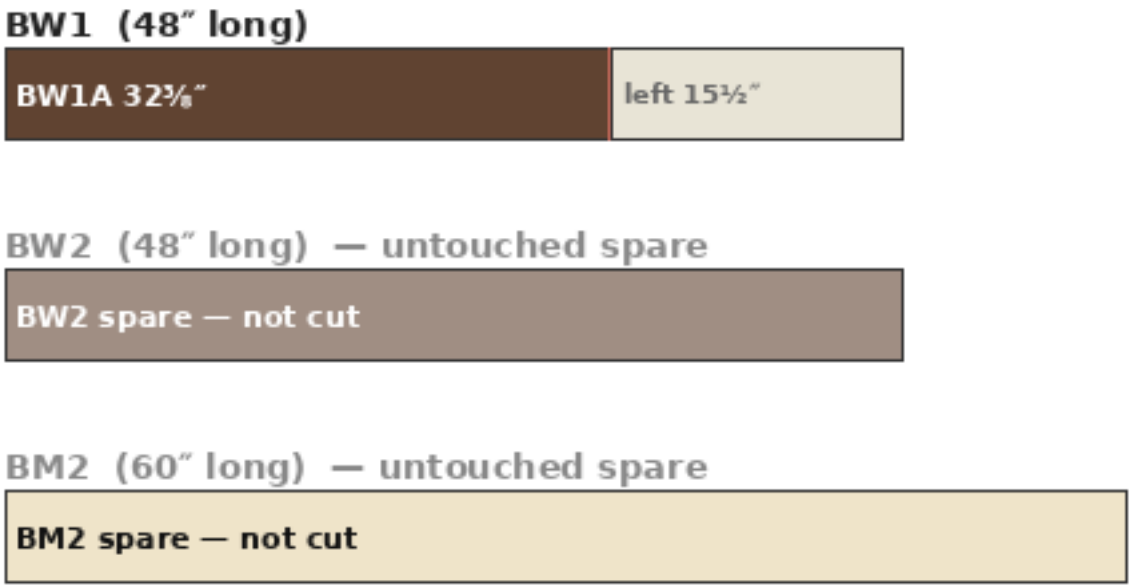
The first operation: crosscut each raw inventory board along its length into the labeled board blanks that get ripped later (the Initial board rips section, which follows immediately, ahead of Panels overview). Boards that start from another board's leftover margin are not raw-stock crosscuts — they appear as the labeled offcuts in the Board Rip Diagram and as a crosscut row (↳) after their parent board's row in the rip cut table.

Raw board	Species	Raw length	Blanks (id = length)	Leftover
BM1	maple	60″	BM1A = 32⅜″	27½″
BC1	cherry	40″	BC1A = 38⅞″	1″
BC2	cherry	40″	BC2A = 32⅜″	7½″
BW1	walnut	48″	BW1A = 32⅜″	15½″
BW2	walnut	48″	— (untouched spare)	—
BM2	maple	60″	— (untouched spare)	—

Crosscut Diagram

Each bar is one raw board drawn to scale by length; segments are the blanks (labeled with id and length inside), reddish stripes are crosscut kerfs (sawdust), and the pale tail is leftover length. A red tail means the blanks overflow the raw board. An untouched spare is shown washed-out and uncut; a board used whole goes straight to the rip diagram. The bars below are the individual primitives, left-aligned (not floated side by side); the combined diagram is also saved in images/ as crosscut-blanks-*.png.





5. Initial board rips (the cut list)

Each row is one physical board at the saw: rip the strips down its width, then crosscut each strip to the piece lengths. The **Board** column shows `id @ width` — the board's width when ripping begins. **Used** is the summed strip width, **Loss** is rip kerf + planer cleanup per strip, **Margin** is the leftover width (negative = the strips do not fit; large = the board is oversized for what it yields).

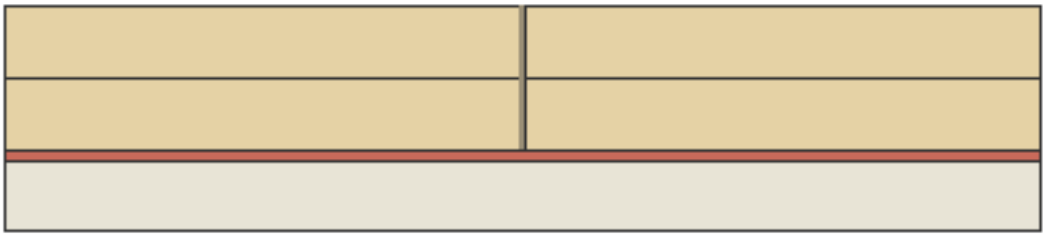
A board whose leftover margin is itself the source of other boards gets a crosscut row (↳) immediately after its own row: it lists, in cut-spec document order, the child boards crosscut off that margin slab, the miter kerf between them, and the slab tail left after the last crosscut. For a crosscut row the Used / Loss / Margin figures are length-axis, not width-axis: Used is the total child length, Loss is the miter kerf consumed, Margin is the slab tail.

Board	Origin	Species	Length	Rips (width × count)	Used	Loss	Margin
BM1A @ 7″	BM1	maple	32⅜″	2¼″ ×2	4½″	⅜″	2⅛″
BC1A @ 6″	BC1	cherry	38⅞″	2¼″ ×2	4½″	⅜″	1⅛″
BC2A @ 5½″	BC2	cherry	32⅜″	1⅛″ ×2	2¼″	⅜″	2⅞″
BW1A @ 5″	BW1	walnut	32⅜″	2¼″ ×2	4½″	⅜″	⅛″

Board Rip Diagram

Each labeled block below is one physical board. Horizontal bands are the strips ripped down the board's width; within a band, the colored cells are the crosscut pieces (with thin kerf gaps between them) and the lightened tail is the offcut. The pale band under the strips is the leftover margin; a gray title marks a board that must be bought. The blocks below are the individual primitives, flowed to fill the page width instead of one tall combined diagram; the combined diagram is also saved as strip-map.png at the plan root and as images/strip-map-*.png, for a single-image copy of the cut diagram.

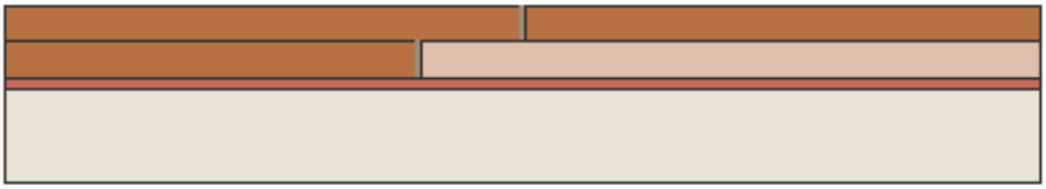
BM1A (32⅜″ x 7″ x 13⁄16″)



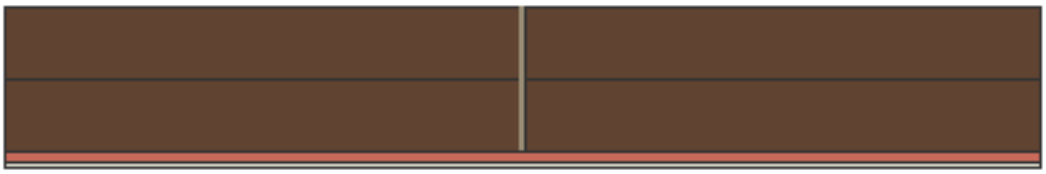
BC1A (38⅞″ x 6″ x 13⁄16″)



BC2A (32⅜″ x 5½″ x 13⁄16″)



BW1A (32⅜″ x 5″ x 13⁄16″)

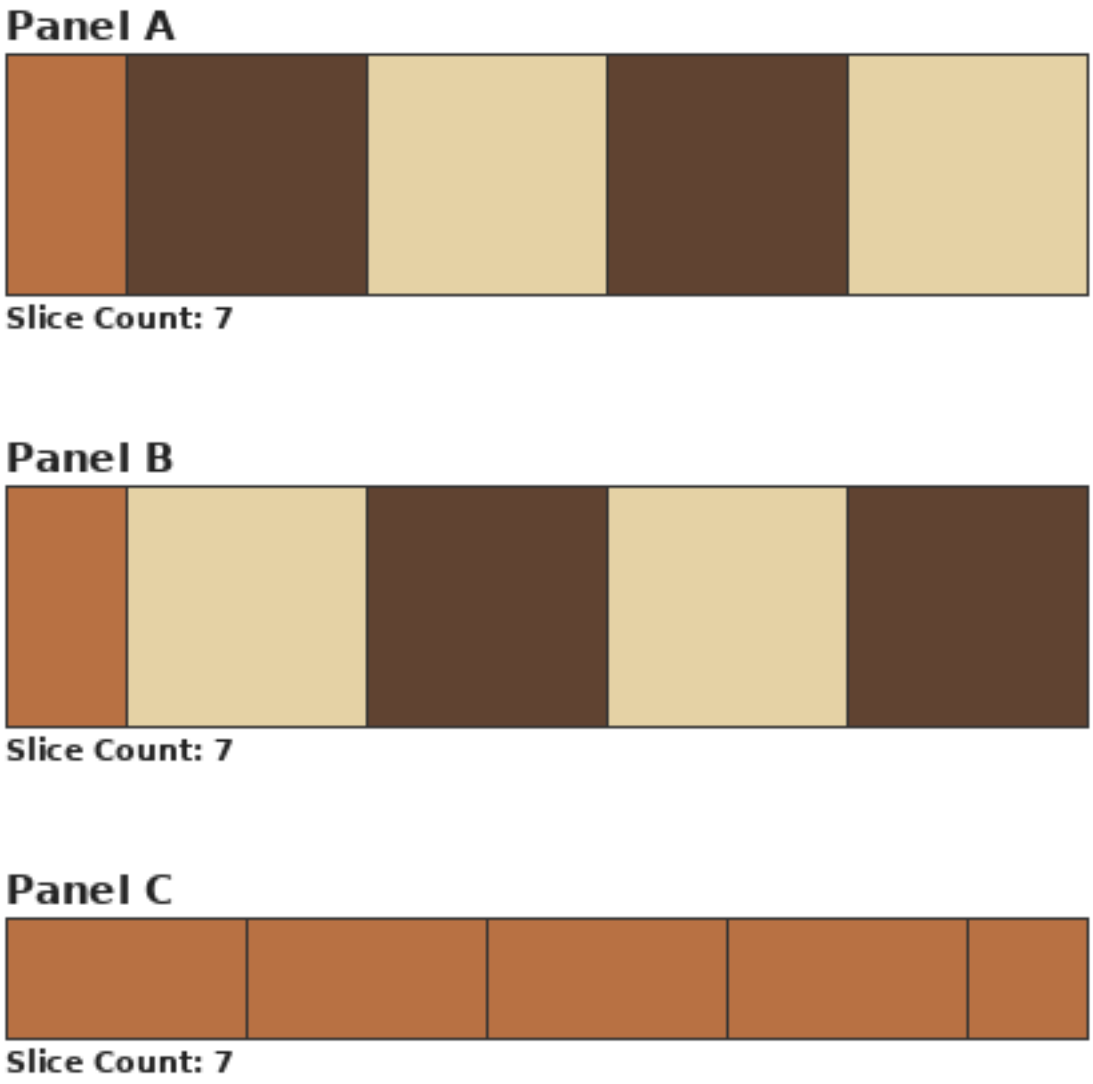


6. Panels overview

Panel	Qty	Length	Slices	Height	Ripped	Sequence (outer → in)
A	1	16⅛″	9	2¼″	—	CB · W · M · W · M
B	1	16⅛″	9	2¼″	—	CB · M · W · M · W
C	1	12⅞″	7	1⅛″	—	CT · CT · CT · CT · CTH

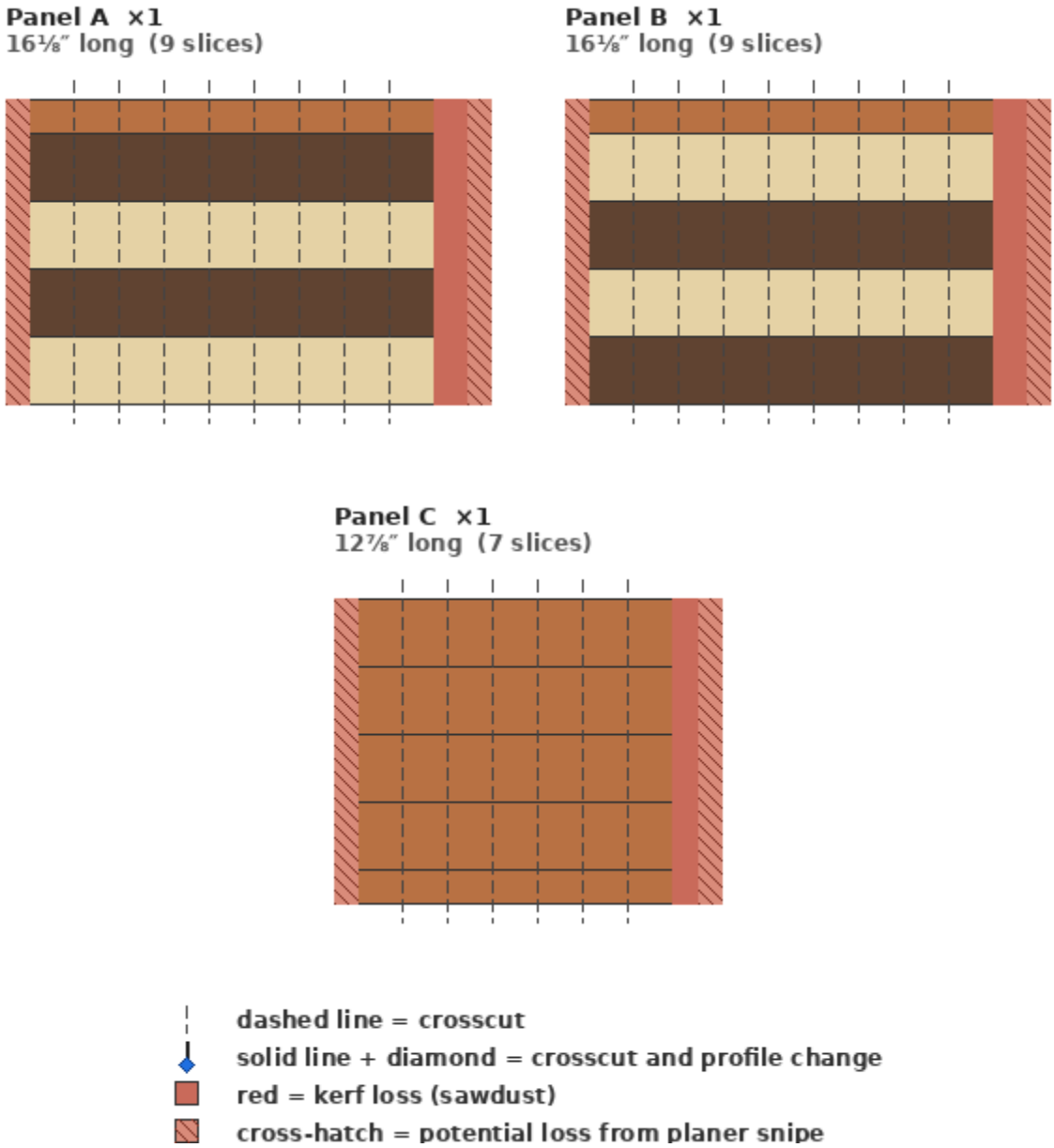
Panel End Grain Diagram

Each panel's end-grain cross-section (the glue-up of component stripes). The panels below are the individual primitives, centered in the PDF; the combined diagram is also saved in images/ as panels-*.png.



Panel Face Diagram

Each panel drawn to scale with its length running left-to-right, wider than tall, centered in the PDF at 75% of its rendered size, with the panel body moved down away from its title. Components are horizontal stripes aligned on the squared edge; profile regions stack left-to-right, so the silhouette steps shorter where trailing components drop out. Dashed lines are the slice crosscuts; the solid red band is total kerf loss, and the cross-hatched ends are the snipe tails (possible loss, depending on the planer). A blue diamond below a solid line marks a crosscut that is also a profile change. The faces below are the individual primitives, on a white background that bleeds into the page; the combined diagram is also saved in images/ as panel-faces-*.png. The legend follows the flow, centered, instead of sitting at the bottom of one tall image.



Slice Fulfillment

Demand is every slice the grid places across all 1 finished board(s). **Supplied** is the slices the panels above produce — snipe lost, ripped panels multiplied by their strip count. **Supply Offset** = supplied − demand: negative is a shortfall, positive is extra (ideally non-negative everywhere).

Slice	Demand	Supplied	Supply Offset
C	4 × 1 = 4	7	+3
A	8 × 1 = 8	9	+1
B	8 × 1 = 8	9	+1

The slice counts supplied are calculated assuming the panels and their quantities are delivered perfectly according to the specs in this section. A panel type may have more than one instance; the ideal panels are always quantity 1.

7. Components

Demand is what the finished pattern needs (counted from the slice grid; length is the panel the component belongs to). **Fulfilled** is how many the cut list supplies, matched longest-demand-first using exact-length pieces first, then longer pieces cut down (a piece shorter than the demand cannot be used). **Shortfall** is the unmet count. **Total Overcut** is the wasted length trimmed off — the sum of (piece length − demand length) across every longer piece that had to be cut down to fill this row (0 when exact-length pieces covered it).

Component	Species	Width	Length	Demand	Fulfilled	Shortfall
CB	cherry	1⅛″	16⅛″	2	2	0
CT	cherry	2¼″	12⅞″	4	4	0
CTH	cherry	1⅛″	12⅞″	1	1	0
M	maple	2¼″	16⅛″	4	4	0
W	walnut	2¼″	16⅛″	4	4	0

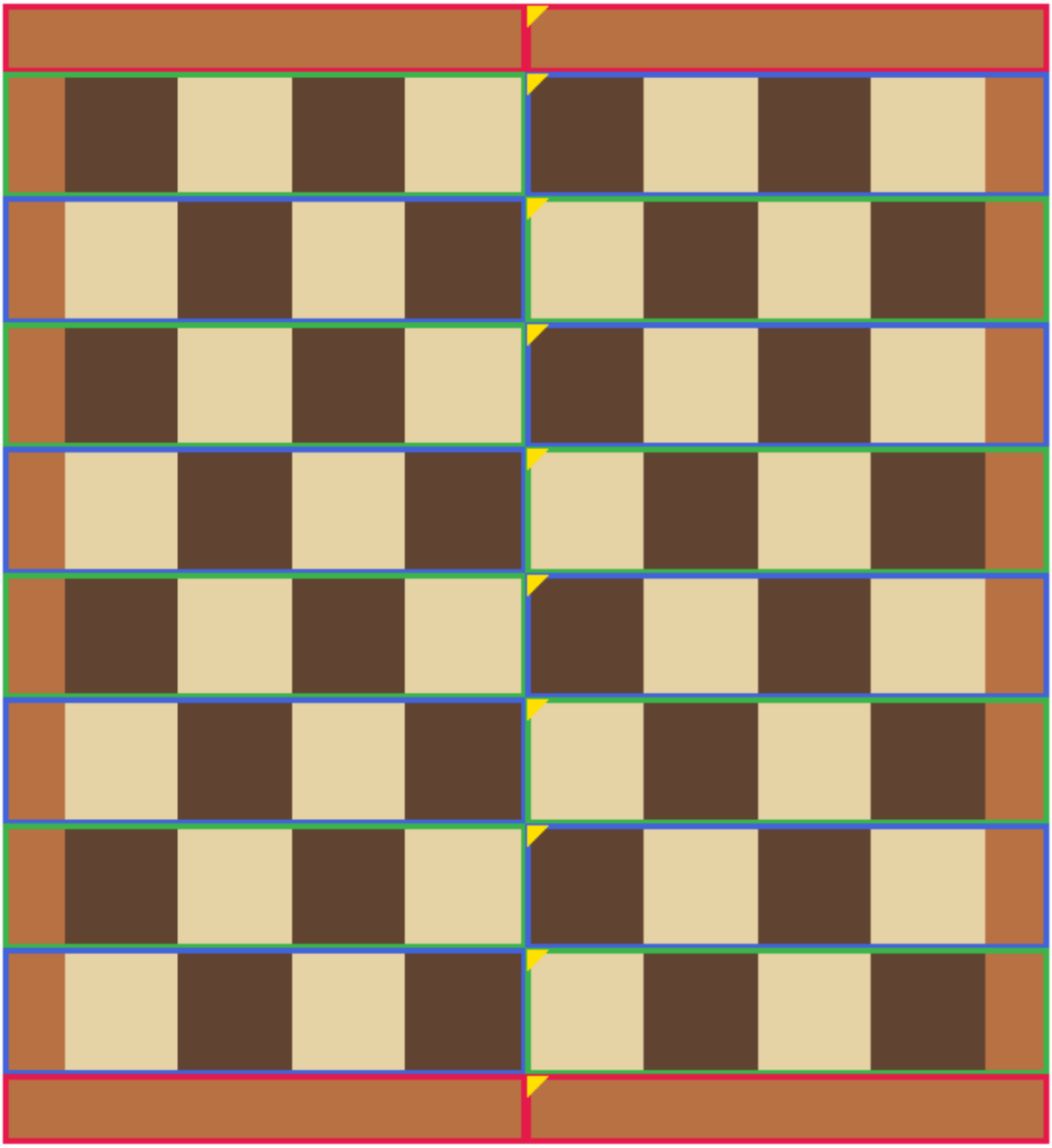
8. Panels (profiles and sequence)

The physical panels this plan builds (from the project's panel specs), grouped by identical shape. Qty is how many of each to glue up.

Panel	Qty	Length	Profiles (slices)	Component sequence
A	1	16⅛″	single: 9	CB · W · M · W · M
B	1	16⅛″	single: 9	CB · M · W · M · W
C	1	12⅞″	single: 7	CT · CT · CT · CT · CTH

9. Slice Grid

The final step: arrange the slices into the finished face. The same grid keyed by slice id — each slice is framed in its legend color so you can see how to arrange the grid cell by cell. A corner triangle marks a slice placed horizontally flipped.



Slice legend

C A B

corner triangle = slice placed horizontally flipped